VIT
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

Reg. Number:

Continuous Assessment Test (CAT)– I JAN 2026

Programme	:	B.Tech(CSE)	Semester	:	WINTER 2025-26
Course Code & Course Title	:	BCSE301L & Software Engineering	Class Number	:	CH2025260502037 CH2025260502033 CH2025260502026
Faculty	:	Dr. R. PRABHAKARAN Dr. ALOK CHAUGAN Prof.S BERIN SHALU	Slot	:	D1+TD1
Duration	:	90 minutes	Max. Mark	:	50

General Instructions:

- Write only your registration number on the question paper in the box provided and do not write other information
- Only non-programmable calculator without storage is permitted

Answer all questions

Q. No	Sub Sec.	Description	Marks	CO	BT Level
1	a)	A Smart Traffic Management System is developed to monitor traffic using sensors and cameras, control signals dynamically, and provide real-time alerts. During development, the software requires frequent updates, does not wear out physically, shows unexpected effects when modified, and needs continuous testing and maintenance. Explain the nature of software and its characteristics. (5 Marks)	10	CO1	K3
	b)	A healthcare organization is developing an AI-based Patient Monitoring System. Describe software engineering as a layered technology with a neat diagram for it. (5 Marks)			
2	a)	You are part of a 6-person software development team building a mobile banking app. The project has been using a traditional waterfall approach, but requirements keep changing due to new RBI regulations and customer feedback. The team is facing delays, low code quality (many bugs found late), poor communication between developers and the business team, and burnout from long hours. The project manager wants to shift to an agile approach and is considering Extreme Programming (XP) as one option because the team is small and co-located. Explain two key values from the Agile Manifesto and two core principles that would directly address the problems in this scenario.(2 marks)	10	CO3	K3
	b)	Describe three specific XP practices, you would recommend implementing first, and explain how each practice helps solve one or more of the current issues.			

	c)	(3 marks) Discuss one potential challenge in adopting XP in this context and how the team could overcome it while staying aligned with agile principles. Conclude with why XP would be suitable compared to sticking with waterfall. (5 marks)			
3	a) b)	A software project team at VIT University consists of students from different departments and cultural backgrounds. During the development of an Online Course Registration System, conflicts arise due to unequal workload distribution, communication gaps, and missed deadlines. Analyze the team dynamics issues present in this scenario. Explain how factors such as roles, leadership, motivation, and conflict affect team performance. (5 Marks) Suggest effective communication and conflict-management strategies that the project manager can adopt to improve collaboration and ensure timely project delivery. (5 Marks)	10	CO2	K4
4	a) b)	The project coordinator at VIT University wants to assess the quality and progress of a Student Feedback Management System being developed by a student team. To ensure timely delivery and acceptable quality, the coordinator decides to apply software metrics and measurement techniques. Illustrate the importance of software metrics in project management. Identify and describe any five project or product metrics suitable for this project. (5 Marks) Discuss how metrics and measurements can be used to monitor schedule, cost, productivity, and software quality in this project. Provide relevant examples. (5 Marks)	10	CO2	K4
5		A new banking web application is analyzed using Function Point Analysis. The following counts are identified: • External Inputs (EI): 12 (average complexity) • External Outputs (EO): 8 (high complexity) • External Inquiries (EQ): 6 (low complexity) • Internal Logical Files (ILF): 5 (average complexity) • External Interface Files (EIF): 4 (low complexity)	10	CO2	K5

a)	Compute the Unadjusted Function Point count (UFP) and Adjusted Function Point (FP), assuming sum of value adjustment factors to be 38. (2 marks) Historical data from similar projects shows: • Productivity = 7 FP per person-month • Average cost per person-month = ₹25,000			
b)	The client requires the application to be delivered in exactly 10 months (including all phases: requirements → deployment). Calculate: • Estimated effort in person-months (2 marks) • Required average team size (2 marks) • Total estimated project cost in ₹ (2 marks) Given a fixed budget of ₹18, 00,000, state whether the project appears feasible and comfortable from a cost and staffing perspective. If there is significant buffer, suggest one reasonable way to use the extra funds productively. (2 marks)			

**********All the best **********

Faculty Signature